INCH-POUND

MIL-I-24768/13
25 February 1992

SUPERSEDING
MIL-P-15035C
10 January 1962
MIL-P-79C
15 June 1961

MILITARY SPECIFICATION SHEET

INSULATION, PLASTIC, LAMINATED, THERMOSETTING, COTTON-FABRIC-BASE, PHENOLIC-RESIN (FBE)

This specification is approved for use by all departments and agencies of the Department of Defense.

The requirements for acquiring the insulation material described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-I-24768.

REQUIREMENTS:

Construction: The material shall consist of a woven cotton fabric substrate impregnated and bonded with a phenolic-resin matrix and processed to meet the requirements of this specification.

Intended use: The laminated material covered by this specification is primarily intended for electrical applications requiring good moisture resistance. This material has low mechanical strength but is suitable for fine machining operations. It possesses fairly good dielectric properties and a medium dielectric loss.

Forms: Sheets, rolled tubes, molded tubes, and rods are recommended forms of laminated material covered by this specification.

Temperature indices: The laminated material covered by this specification has a mechanical strength test index of 85 and an electrical strength test index of 85 for thicknesses up to 0.062 inch, and a mechanical strength test index of 125 and an electrical strength test index of 115 for thicknesses greater than 0.062 inch.

Dimensions and part numbers:

Thickness of sheets: The thickness of laminated sheets, permissible variations, and the applicable part number shall be as specified in table I.

AMSC N/A FSC 5970

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T01:59Z
Check the source to verify that this is the current version before use.

TABLE I. *Thickness of laminated sheets.*

Part number	Nominal thickness (inches)	Permissible variations (± inch)
M24768/13-S-1	0.015	0.0035
M24768/13-S-2	0.020	0.004
M24768/13-S-3	0.025	0.0045
M24768/13-S-4	0.031	0.005
M24768/13-S-5	0.047	0.0055
M24768/13-S-6	0.062	0.006
M24768/13-S-7	0.094	0.007
M24768/13-S-8	0.125	0.008
M24768/13-S-9	0.156	0.009
M24768/13-S-10	0.188	0.010
M24768/13-S-11	0.219	0.011
M24768/13-S-12	0.250	0.012
M24768/13-S-13	0.312	0.015
M24768/13-S-14	0.375	0.017
M24768/13-S-15	0.438	0.019
M24768/13-S-16	0.500	0.021
M24768/13-S-17	0.625	0.024
M24768/13-S-18	0.750	0.027
M24768/13-S-19	0.875	0.030
M24768/13-S-20	1.000	0.033
M24768/13-S-21	1.125	0.035
M24768/13-S-22	1.250	0.037
M24768/13-S-23	1.375	0.039
M24768/13-S-24	1.500	0.041
M24768/13-S-25	1.625	0.043
M24768/13-S-26	1.750	0.045
M24768/13-S-27	1.875	0.047
M24768/13-S-28	2.000	0.049
M24768/13-S-29	_[1]	_[2]

[1]For part numbers of sheets not listed in this table, the thickness shall be included after the part number. Example of this for a 0.8125 inch thick sheet is: M24768/13-S-29-0.8125.

[2]On sheets of nominal thickness not listed in this table, the permissible variations shall be the same as for the next greater thickness.

Diameter of rolled round tubes: The range of sizes for rolled round tubes shall be as specified in table II. The inside diameter and outside diameter shall be included in the part number. An example of a part number for a rolled round tube with an inside diameter of 0.188 inch and an outside diameter of 0.250 inch is: M24768/13-TRR-0.188/0.250. The wall thickness tolerances for rolled tubes with an inside diameter up to 4 inches shall be as specified in table III.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T01:59Z
Check the source to verify that this is the current version before use.

TABLE II. *Range of sizes for rolled round tubes.*

Inside diameter (inches)		Outside diameter (inches)		Wall thickness (inch)		Maximum ratio of wall thickness to inside diameter
Minimum	Maximum	Minimum	Maximum	Minimum	Maximum	
0.188	48.000	0.250	50.000	0.031	1.000	1/2

TABLE III. *Rolled round tube wall thickness tolerances.*

Wall thickness (inch)	Tolerances (± inch)	
	Inside diameter (inches)	
	0.188–0.500	0.501–Max.
0.031 to 0.061	0.010	0.008
0.062 to 0.124	0.011	0.009
0.125 to 0.249	0.013	0.011
0.250 to 0.500	0.015	0.013

Diameter of molded round tubes: The range of sizes of molded round tubes shall be as specified in table IV. The inside diameter and outside diameter shall be included in the part number. An example of a part number for a molded round tube with an inside diameter of 0.125 inch and an outside diameter of 0.188 inch is: M24768/13-TMR-0.125/0.188. The wall thickness tolerances for molded round tubes with an inside diameter up to 3.875 inches shall be as specified in table V.

TABLE IV. *Range of sizes for molded round tubes.*

Inside diameter (inches)		Outside diameter (inches)		Wall thickness (inch)		Maximum ratio of wall thickness to inside diameter
Minimum	Maximum	Minimum	Maximum	Minimum	Maximum	
0.125	3.875	0.188	4.000	0.031	1.000	1/2

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T01:59Z
Check the source to verify that this is the current version before use.

TABLE V. *Molded round tube wall thickness tolerances.*

Wall thickness (inch)	Tolerances (± inch)		
	Inside diameter (inches)		
	0.125–0.250	0.251–0.500	0.501–Max.
0.031 to 0.061	0.008	0.008	0.008
0.062 to 0.125	0.11	0.011	0.011
0.126 to 0.250		0.015	0.011
0.251 to 0.500			0.013

Diameter of rods: The range of sizes and tolerances for rods shall be as specified in table VI. The outside diameter of the rod shall be included in the part number. An example of a part number for a rod with an outside diameter of 0.188 inch is: M24768/13-R-0.188.

TABLE VI. *Size and tolerances for rods.*

Diameter (inches)	Tolerances (± inch)
0.125 to 1.999	0.005
2.000 to 4.000	0.008

Performance requirements: Performance requirements for laminated materials shall be as specified in tables VII, VIII, IX, X, and XI.

The following performance requirements do not apply to this plastic laminate:

a. Volume resistivity
b. Arc resistance
c. Flammability
d. Silicone content
e. Thermal endurance
f. Tracking resistance.

Testing: The laminated material covered by this specification shall be tested under first article inspection requirements.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T01:59Z
Check the source to verify that this is the current version before use.

TABLE VII. *Performance requirements for sheets.*

Requirement	Cond.	Test para.	Unit	Thickness (inches)[1]									
				0.031	0.062	0.094	0.125	0.188	0.250	0.500	0.750	1.000	1.001 – Max.
Dielectric breakdown voltage[2]	A D-48/50	4.6.4	Min. kV	40.0 3.0	40.0 3.0	40.0 3.0	40.0 3.0	40.0 3.0	40.0 3.0	40.0 3.0	40.0 3.0	40.0 3.0	30.0 3.0
Dielectric constant	A D-24/23	4.6.6	Max.	5.80	5.80	5.80	5.80 6.00	5.80 6.00	5.80 6.00	5.80 6.00	5.80 6.00	5.80 6.00	5.80 6.00
Dissipation factor	A D-24/23	4.6.6	Max.	0.055	0.055	0.055	0.055 0.070	0.055 0.070	0.055 0.070	0.055 0.070	0.055 0.070	0.055 0.070	0.055 0.070
Volume resistivity	C-96/35/90	4.6.7	Min. megohm cm										
Surface resistance	C-96/35/90	4.6.8	Min. megohm				0.5						
Arc resistance	D-48/50	4.6.9	Min. sec.										
Tracking resistance	A	4.6.10	Min. minutes										
Impact strength[2] Lengthwise: Crosswise:	E-48/50	4.6.11	Min. ft-lbs. per inch				1.25 1.00	1.25 1.00	1.25 1.00	1.25 1.00	1.25 1.00	1.25 1.00	1.25 1.00
Flexural strength Lengthwise: Crosswise:	A	4.6.12	Min. psi	16,000 14,000	16,000 14,000	16,000 14,000	16,000 14,000	16,000 14,000	16,000 14,000	15,000 13,000	15,000 13,000	14,500 13,000	14,500 13,000

See footnotes at end of table.

Source: https://assist.dla.mil - Downloaded: 2026-10-20T01:59Z
Check the source to verify that this is the current version before use.

TABLE VII. *Performance requirements for sheets – Continued.*

Requirement	Cond.	Test para.	Unit	Thickness (inches)[1]									
				0.031	0.062	0.094	0.125	0.188	0.250	0.500	0.750	1.000	1.001 – Max.
Thermal endurance Flexural strength:		4.6.13	Min. psi										
Bonding strength[2]	A D-48/50	4.6.15	Min. pounds							1,600 1,500	1,600 1,500	1,600 1,500	1,600 1,500
Water absorption	D_1-24/23	4.6.17	Max. percent	4.00	1.95	1.55	1.30	1.00	0.95	0.70	0.60	0.55	0.55
Silicone content	E-168/185	4.6.18	Max. ppm										
Flame resistance													
Ignition time:	A	4.6.19	Min. sec.										
Burning time:			Max. sec.										

[1]For intermediate thicknesses, the value for the next smaller thickness shall apply.
[2]Maximum thickness tested shall be 2.000 inches.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T01:59Z
Check the source to verify that this is the current version before use.

TABLE VIII. *Performance requirements for rolled tubes.*

Requirement	Cond.	Test para.	Unit	Inside diameter (inches)	Wall thickness (inch)				
					0.031–0.047	0.048–0.062	0.063–0.094	0.095–0.125	0.126–0.188
Dielectric strength	A	4.6.5	Min. volts per mil	0.188–0.499	120	120	140	140	120
				0.500–8.000		120	140	140	120
Compressive strength[1]	A	4.6.14	Min. psi	0.188–0.499			13,000	13,000	13,000
				0.500–8.000			13,000	13,000	13,000
Specific gravity	A	4.6.16	Min.	0.188–0.499	1.12	1.12	1.12	1.12	1.12
				0.500–8.000	1.14	1.14	1.14	1.14	1.14
Water absorption	D_1-24/23	4.6.17	Max. percent	0.188–8.000	7.5	7.5	5.0	3.0	2.5

TABLE VIII. *Performance requirements for rolled tubes – Continued.*

Requirement	Cond.	Test para.	Unit	Inside diameter (inches)	Wall thickness (inch)				
					0.189–0.250	0.251–0.375	0.376–0.500	0.501–0.750	0.751–1.000
Dielectric strength	A	4.6.5	Min. volts per mil	0.188–0.499	120	85	85	70	60
				0.500–8.000	120	85	85	70	60
Compressive strength[1]	A	4.6.14	Min. psi	0.125–0.499	13,000	13,000	13,000	13,000	13,000
				0.500–8.000	13,000	13,000	13,000	13,000	13,000
Specific gravity	A	4.6.16	Min.	0.125–0.499	1.12	1.12	1.12	1.12	1.12
				0.500–8.000	1.14	1.14	1.14	1.14	1.14
Water absorption	D_1-24/23	4.6.17	Max. percent	0.188–8.000	1.9	1.5	1.2	0.9	0.9

[1]Values apply to inside diameters of 0.250 inch and over and to outside diameters of 2.000 inches and less.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T01:59Z
Check the source to verify that this is the current version before use.

TABLE IX. *Performance requirements for molded tubes.*

Requirement	Cond.	Test para.	Unit	Inside diameter (inches)	Wall thickness (inch)							
					0.031–0.061	0.062–0.093	0.094–0.124	0.125–0.187	0.188–0.249	0.250–0.499	0.500–0.999	1.000
Compressive strength	A	4.6.14	Min. psi	0.250 to 3.875		19,000	19,000	19,000	19,000	19,000	19,000	19,000
Specific gravity	A	4.6.16	Min.	0.125 to 3.875	1.25	1.25	1.25	1.25	1.25	1.25	1.25	1.25
Water absorption	D_1-24/23	4.6.17	Max. percent	0.125 to 3.875	4.5	2.2	1.8	1.5	1.3	1.2	1.0	0.9

TABLE X. *Dielectric strength requirements for molded tubes.*

Requirement	Cond.	Test para.	Unit	Inside diameter (inches)	Wall thickness (inch)			
					0.062	0.063–0.125	0.126–0.250	0.251–0.500
Dielectric strength	A	4.6.5	Min. volts per mil	0.125-3.875	150[1]	175	125	90

[1]The mold seam has a more pronounced effect on the dielectric strength of tubes having a wall thickness of 0.062 inch than on the dielectric strength of tubes having a wall thickness greater than 0.062 inch.

TABLE XI. *Performance requirements for rods.*

Requirement	Cond.	Test para.	Unit	Diameter (inches)				
				0.125–0.249	0.250–0.499	0.500–0.999	1.000	1.001–2.000
Flextural strength	A	4.6.12	Min. psi	12,000	12,000	12,000	12,000	
Compressive strength	A	4.6.14	Min. psi	20,000	20,000	20,000	20,000	20,000
Specific gravity	A	4.6.16	Min.	1.26	1.26	1.26	1.26	1.26
Water absorption	D_1-24/23	4.6.17	Max. percent	2.2[1]	1.4	1.1	1.0	1.0

[1]This value applies to rods having a diameter of 0.188 inch.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T01:59Z
Check the source to verify that this is the current version before use.

Custodians:
Army – ER
Navy – SH
Air Force – 20

Review Activities:
Army – MI, AR, EA
Navy – EC, OS
Air Force – 85

User Activities:
Army – ME
Navy – MC, AS

Preparing Activity:
Navy – SH
(Project 5970-1015-13)

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T01:59Z
Check the source to verify that this is the current version before use.